

Data Limitations

Multi-Branch Retail Performance & Operations Intelligence

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Tools: Excel • MySQL • Power BI • Google Docs • Google Slides • GitHub

This document is the master limitations register for the project. It should be updated only when later SQL analysis verifies or clarifies a limitation. Read alongside the Business Insights and Recommendations documents so that conclusions are never taken beyond what the dataset can reliably support.

Source-Data Limitations

No explicit product cost / COGS

- Selling-price information is available but explicit product cost/COGS is not.
- True gross profit, gross margin, net profit and accounting profitability therefore cannot be derived reliably.
- Revenue/transaction value must never be relabeled as profit.

Customer signup-date inconsistency

- 120,020 orders have order_date earlier than the associated signup_date.
- Signup_date is unreliable for tenure, acquisition cohorts and pre/post-signup behavior.
- Orders should be retained and signup_date excluded from those analyses.

No return-date field

- True return-month trends, return timing and time-to-return cannot be calculated.
- Time grouping of returns must be described as association with the original order month/date.

Multiple return records can reference one order item

- Counting return records as unique returned items can inflate return rates.
- Use COUNT(DISTINCT order_item_id) and appropriate refund aggregation.

Incomplete / partial final time period

- The final period may not be comparable with complete earlier periods.
- Apparent end-of-series declines can be exaggerated.
- Use like-for-like comparisons.

No payment method / status

- The payments data available to the project does not contain payment method or payment status, so those analyses are not claimed.

Shipment status is limited to SHIPPED, DELIVERED and LATE

- Detailed carrier, route, warehouse and SLA attributes are unavailable.

Dataset is historical / static

- The dataset is historical/static rather than a live operational feed.

Analytical Caveats & Interpretation Rules

- ❖ Use `order_items.qty × order_items.price` for historical sales calculations; `products.price` is the master/product price while `order_items.price` is the transaction-level selling price.
- ❖ Promotion discount application is not fully specified. Do not subtract `promotions.discount` again unless the transaction-price semantics are confirmed, because this could double-count the discount.
- ❖ Promotion comparisons are associational, not causal. Use 'associated with' rather than 'caused by'.
- ❖ Payment-to-order-value differences require investigation and are not automatically payment errors.
- ❖ Gross transaction value is not profit; use gross sales, transaction value or revenue.
- ❖ Refunds must be aggregated at the correct order-item grain because an item can have multiple return/refund records.
- ❖ Return rate requires a defined denominator; for order-item return rate use distinct returned `order_item_id` divided by total order-item records.
- ❖ Shipment/return relationships are descriptive. A higher return rate among shipment statuses does not prove shipment status caused the return.
- ❖ Order-date-based return trends are associations with the original order period, not actual return timing.
- ❖ Absolute late-shipment counts should not alone define the worst city; normalized rates are more appropriate for fair comparison.

Supported vs Unsupported Claims

Supported metrics

Gross sales/transaction value, order volume, units sold, AOV, purchase frequency, store/product/category/supplier sales distributions, descriptive promotion comparisons, payment distributions and reconciliation, shipment-status distributions, distinct order-item return rates and aggregated refunds.

Not supported without additional data

Gross profit, net profit, profit margin or true profitability.

Not supported

- ❖ Reliable customer tenure or acquisition cohorts based on `signup_date`.
- ❖ Actual return-date trends or time-to-return.
- ❖ Causal claims that promotions increase sales.
- ❖ Causal claims that shipment status causes returns.
- ❖ Subtracting `promotions.discount` again to create post-discount net sales without confirmed price semantics.

Preferred Terminology

Use	Avoid
Gross sales / transaction value / revenue	Profit / gross profit / profit margin
Promotion-associated sales	Sales caused by promotion
Distinct returned order items	Return records treated as unique returned items
Returned items associated with original order month	Actual monthly returns
Payment difference / reconciliation difference	Payment error
Higher return rate among late-shipment orders	Late shipment caused the return

Appendix — Validation Notes

- ❖ Late Shipments: SQL 99,855; Power BI displayed 100K due to display-unit rounding; underlying value matched.
- ❖ Total Refunds: SQL 75,962,300.00; Power BI displayed 75.96M; underlying value matched.
- ❖ Returned Items: SQL 29,251; Power BI displayed 29K; underlying value matched.
- ❖ Customer Value Distribution uses an independent numeric sort column.
- ❖ City-level shipment and return visuals required correct relationship/filter propagation.
- ❖ Dashboard 3 KPI was renamed from Total Products Sold to Total Products because it counts distinct product IDs.
- ❖ Payment method/status analysis is not claimed because those fields are unavailable.